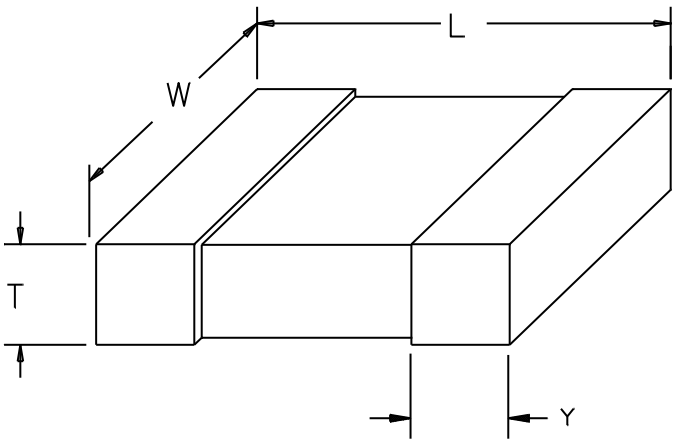
INCH-POUND
MIL-PRF-55681/13
12 July 2016

PERFORMANCE SPECIFICATION SHEET

CAPACITOR, CHIP, MULTIPLE LAYER, FIXED, CERAMIC DIELECTRIC, ESTABLISHED RELIABILITY AND NON-ESTABLISHED RELIABILITY, STYLE CDR37

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-55681.



Dimensions			
L ± .004	W ± .004	T Max.	Y ± .006
.040	.020	.024	.010

inches	mm
.004	0.10
.006	0.15
.010	0.25
.020	0.51
.024	0.61
.040	1.02

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Dimensions and tolerances are for terminated chips.
4. For solder termination finishes, add .020 inch (0.51 mm) to the positive length tolerance and .015 inch (0.38 mm) to the positive width and thickness tolerances. The increase in dimension applies to the solder coating thickness only.

FIGURE 1. Style CDR37 capacitors.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Rated temperature and voltage-temperature limits: BP, BX, and BR as specified in MIL-PRF-55681.

Capacitance value: See table I.

Capacitance tolerance:

Symbol	Capacitance tolerance (±)
C	0.25 pF
D	0.5 pF
F	1 percent
G	2 percent
J	5 percent
K	10 percent
M	20 percent

Rated voltage:

Symbol	Rated voltage (V_{dc})
V	4
W	6.3
X	10
Y	16
Z	25
A	50
B	100

Operating temperature range: -55° C to +125° C.

Termination finish: M, U, and Z as specified in MIL-PRF-55681.

Product level designator: Non-established reliability (C) and established reliability (M, P, R, and S) as specified in MIL-PRF-55681.

Marking: In accordance with MIL-PRF-55681.

TABLE I. Style CDR37 capacitor characteristics.

Part or Identifying Number (PIN) 1/	Capacitance (pF)	Capacitance tolerance	Rated temperature and voltage-temperature limits	Rated voltage (V_{dc}) 2/
CDR37BP0R5- - - -	.5	C, D	BP	100
CDR37BPR75- - - -	.75	C, D	BP	100
CDR37BP1R0- - - -	1	C, D	BP	100
CDR37BP1R2- - - -	1.2	C, D	BP	100
CDR37BP1R5- - - -	1.5	C, D	BP	100
CDR37BP1R8- - - -	1.8	C, D	BP	100
CDR37BP2R2- - - -	2.2	C, D	BP	100
CDR37BP2R4- - - -	2.4	C, D	BP	100
CDR37BP2R7- - - -	2.7	C, D	BP	100
CDR37BP3R0- - - -	3	C, D	BP	100

See footnotes at end of table.

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TABLE I. Style CDR37 capacitor characteristics - Continued.

Part or Identifying Number (PIN) 1/	Capacitance (pF)	Capacitance tolerance	Rated temperature and voltage-temperature limits	Rated voltage (V_{dc}) 2/
CDR37BP3R3- - - -	3.3	C, D	BP	100
CDR37BP3R6- - - -	3.6	C, D	BP	100
CDR37BP3R9- - - -	3.9	C, D	BP	100
CDR37BP4R7- - - -	4.7	C, D	BP	100
CDR37BP5R1- - - -	5.1	C, D	BP	100
CDR37BP5R6- - - -	5.6	C, D	BP	100
CDR37BP6R2- - - -	6.2	C, D	BP	100
CDR37BP6R8- - - -	6.8	C, D	BP	100
CDR37BP7R5- - - -	7.5	C, D	BP	100
CDR37BP8R2- - - -	8.2	C, D	BP	100
CDR37BP9R1- - - -	9.1	C, D	BP	100
CDR37BP100- - - -	10	F, G, J	BP	100
CDR37BP110- - - -	11	F, G, J	BP	100
CDR37BP120- - - -	12	F, G, J	BP	100
CDR37BP130- - - -	13	F, G, J	BP	100
CDR37BP150- - - -	15	F, G, J	BP	100
CDR37BP160- - - -	16	F, G, J	BP	100
CDR37BP180- - - -	18	F, G, J	BP	100
CDR37BP200- - - -	20	F, G, J	BP	100
CDR37BP220- - - -	22	F, G, J	BP	100
CDR37BP240- - - -	24	F, G, J	BP	100
CDR37BP270- - - -	27	F, G, J	BP	100
CDR37BP300- - - -	30	F, G, J	BP	100
CDR37BP330- - - -	33	F, G, J	BP	100
CDR37BP360- - - -	36	F, G, J	BP	100
CDR37BP390- - - -	39	F, G, J	BP	100
CDR37BP430- - - -	43	F, G, J	BP	100
CDR37BP470- - - -	47	F, G, J	BP	100
CDR37BP510- - - -	51	F, G, J	BP	100
CDR37BP560- - - -	56	F, G, J	BP	100
CDR37BP620- - - -	62	F, G, J	BP	100
CDR37BP680- - - -	68	F, G, J	BP	100
CDR37BP750- - - -	75	F, G, J	BP	100
CDR37BP820- - - -	82	F, G, J	BP	100
CDR37BP910- - - -	91	F, G, J	BP	100
CDR37BP101- - - -	100	F, G, J	BP	100
CDR37BP121- - - -	120	F, G, J	BP	100
CDR37BP151- - - -	150	F, G, J	BP	100
CDR37BP181- - - -	180	F, G, J	BP	100
CDR37BP221- - - -	220	F, G, J	BP	50
CDR37BP271- - - -	270	F, G, J	BP	16
CDR37BP331- - - -	330	F, G, J	BP	16
CDR37BX101- - - -	100	J, K M	BX	100
CDR37BX121- - - -	120	J, K M	BX	100
CDR37BX151- - - -	150	J, K M	BX	100
CDR37BX181- - - -	180	J, K M	BX	100
CDR37BX221- - - -	220	J, K M	BX	100
CDR37BX271- - - -	270	J, K M	BX	100
CDR37BX331- - - -	330	J, K M	BX	100
CDR37BX391- - - -	390	J, K M	BX	100

See footnotes at end of table.

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TABLE I. Style CDR37 capacitor characteristics - Continued.

Part or Identifying Number (PIN) 1/	Capacitance (pF)	Capacitance tolerance	Rated temperature and voltage-temperature limits	Rated voltage (V_{dc}) 2/
CDR37BX471- - - -	470	J, K M	BX	100
CDR37BX561- - - -	560	J, K M	BX	100
CDR37BX681- - - -	680	J, K M	BX	100
CDR37BX821- - - -	820	J, K M	BX	50
CDR37BX102- - - -	1,000	J, K M	BX	50
CDR37BX122- - - -	1,200	J, K M	BX	50
CDR37BX152- - - -	1,500	J, K M	BX	50
CDR37BX182- - - -	1,800	J, K M	BX	50
CDR37BX222- - - -	2,200	J, K M	BX	25
CDR37BX272- - - -	2,700	J, K M	BX	25
CDR37BX332- - - -	3,300	J, K M	BX	25
CDR37BX392- - - -	3,900	J, K M	BX	25
CDR37BX472- - - -	4,700	J, K M	BX	16
CDR37BX562- - - -	5,600	J, K M	BX	16
CDR37BX682- - - -	6,800	J, K M	BX	16
CDR37BX822- - - -	8,200	J, K M	BX	16
CDR37BR101- - - -	100	J, K M	BR	100
CDR37BR121- - - -	120	J, K M	BR	100
CDR37BR151- - - -	150	J, K M	BR	100
CDR37BR181- - - -	180	J, K M	BR	100
CDR37BR221- - - -	220	J, K M	BR	100
CDR37BR271- - - -	270	J, K M	BR	100
CDR37BR331- - - -	330	J, K M	BR	100
CDR37BR391- - - -	390	J, K M	BR	100
CDR37BR471- - - -	470	J, K M	BR	100
CDR37BR561- - - -	560	J, K M	BR	100
CDR37BR681- - - -	680	J, K M	BR	100
CDR37BR821- - - -	820	J, K M	BR	100
CDR37BR102- - - -	1,000	J, K M	BR	100
CDR37BR122- - - -	1,200	J, K M	BR	50
CDR37BR152- - - -	1,500	J, K M	BR	50
CDR37BR182- - - -	1,800	J, K M	BR	50
CDR37BR222- - - -	2,200	J, K M	BR	50
CDR37BR272- - - -	2,700	J, K M	BR	50
CDR37BR332- - - -	3,300	J, K M	BR	50
CDR37BR392- - - -	3,900	J, K M	BR	50
CDR37BR472- - - -	4,700	J, K M	BR	25
CDR37BR562- - - -	5,600	J, K M	BR	16
CDR37BR682- - - -	6,800	J, K M	BR	16
CDR37BR822- - - -	8,200	J, K M	BR	16
CDR37BR103- - - -	10,000	J, K M	BR	16

1/ The complete PIN shall include additional symbols to indicate voltage, capacitance tolerance, termination, and product level.

2/ This is the maximum rated voltage available. All lower voltage ratings are also available.

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Custodians:
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Navy - EC
Air Force – 85
DLA – CC

Preparing activity:
DLA - CC

(Project 5910-2013-013)

Review activities:
Navy - AS, MC, OS, SH
Air Force - 19, 99
Other - MDA

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